MADE BY

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CBSE CLASS 12

BUSINESS STUDIES

PREMIUM NOTES

CHAPTER 8

Controlling

NCERT BASED | EXAM FOCUSED | EASY LANGUAGE

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Chapter Snapshot

#	Question	Answer
1	What is this chapter about?	How managers check whether everything is going according to plan — and take corrective action when it is not.
2	Why important?	Without controlling, even perfect plans fail because no one checks if the plan is actually being executed.
3	Real-life use	Every CBSE board exam result is a control mechanism — teachers compare actual marks against target, then decide whether to re-teach.
4	Key concept	Planning and Controlling are twins of management — they cannot exist without each other.
5	Most tested topics	Steps in controlling process (5 steps), Planning vs Controlling relationship, Importance of controlling, Techniques.
6	Exam weight	High — steps in controlling + importance + planning-controlling relationship are regular 4-5 mark questions.
7	Difficulty level	Low-Medium — mostly logical and straightforward. Techniques section needs memorisation.
8	Memory anchors	5 steps: S-M-C-A-T. Techniques: Traditional (P-S-B-B) + Modern (R-R-R-M-P-M).
9	Revision time	First read 1.5 hrs. Second revision 45 min. Final 20 min.
10	One-line summary	Controlling = measuring actual performance → comparing with standards → finding deviations → taking corrective action to bring the organisation back on track.

Section 1 Big Picture of the Chapter

Imagine Smit Sir plans to get 90% of his students to score above 80 marks in Business Studies.

He recruits 5 teachers (Staffing), creates a batch schedule (Organising), and starts the teaching year (Directing). Three months later — mid-term exams happen.

Results come in: Only **60% of students scored above 80. Target = 90%. Actual = 60%. Deviation = 30%.**

Now what? Does Smit Sir just accept this and move on? NO. He:

- **Checks:** Which topics had the most errors? Which batches underperformed? Which teachers need support?
- **Analyses:** Are students weak in theory or application? Was the schedule too fast?
- **Acts:** Adds extra revision sessions, revises the teaching approach, gives targeted practice tests.

That entire process — measuring, comparing, finding the gap, and fixing it — is CONTROLLING.

Controlling is the **last function of management** — but also the link back to planning. It asks: **'Are we on track? If not, what do we fix?'**

Planning and Controlling are called the **'twins of management'** because they are inseparable. A plan without control is a wish. Control without a plan has no standard to measure against.

TEACHER TIP — SMIT SIR

Classroom hook: Ask: 'You planned to score 90% in this exam. Exam result = 65%. What do you do?' Every answer they give — study harder, change method, ask for help, identify weak topics — IS the controlling process. Connect the exam analogy to every step in the chapter.

Section 2 Learning Objectives

After studying this chapter, you will be able to:

1. Define controlling in CBSE language and explain its features.
2. Explain the importance of controlling in an organisation.
3. Explain the close relationship between planning and controlling.
4. Describe the 5-step controlling process in sequence with examples.
5. Explain the concept of 'deviation' and 'critical deviation'.
6. Describe all 4 traditional techniques of managerial control.
7. Describe all 6 modern techniques of managerial control.
8. Distinguish between planning and controlling.
9. Write structured board exam answers for all controlling topics.
10. Identify controlling concepts from case-study questions.

Section 3 Chapter Map

Topic No.	Topic Name	What You'll Learn
1	Meaning of Controlling	Definition + 5 Features
2	Importance of Controlling	7 reasons why controlling is essential
3	Planning and Controlling Relationship	Why they are called 'twins of management'
4	Steps in Controlling Process	5 sequential steps: S-M-C-A-T
5	Techniques of Control — Traditional	Personal Observation, Statistical Reports, Break-even Analysis, Budgetary Control
6	Techniques of Control — Modern	ROI, Ratio Analysis, Responsibility Accounting, Management Audit, PERT & CPM, MIS

MEMORY TRICK

Anchor: WHAT controlling is → WHY it matters → HOW it relates to planning → HOW TO do it (5 steps) → TOOLS to do it (techniques).

Five steps (S-M-C-A-T): Setting standards → Measuring actual → Comparing → Analysing deviation → Taking corrective action.

Recall: "Students Must Compare Actual Tests."

Section 4 Key Terms Before Starting

Term	Simple Meaning	One-line Example
Controlling	Checking whether all is going as planned — and taking corrective action if not.	Comparing exam scores against target, then deciding remedial action.
Standard / Benchmark	A pre-set goal or target against which actual performance is measured.	Smit Sir sets 90% students above 80 marks as the standard.
Actual performance	What actually happened — the real output/result achieved.	Only 60% students scored above 80 — actual performance.
Deviation	The difference between actual performance and the set standard.	90% - 60% = 30% = the deviation.
Critical deviation	A deviation that is significant enough to demand immediate corrective action.	A 30% deviation from target is critical — needs immediate intervention.
Minor deviation	A small deviation that can be ignored or monitored without immediate action.	A 1-2% deviation from target — acceptable variance.
Corrective action	Action taken to bring actual performance in line with standards.	Adding extra revision sessions, changing teaching methods.
Management by Exception	Focus on critical deviations only — let minor deviations go unaddressed.	Manager only investigates deviations above ±10%; smaller ones are ignored.
Budgetary control	Comparing actual financial performance with the budget to identify variances.	Actual sales ₹45 lakh vs Budget ₹50 lakh → ₹5 lakh unfavourable variance.
Break-even analysis	Finding the level of sales at which total revenue = total cost (no profit, no loss).	At 5,000 units, revenue = cost = ₹10 lakh — that is the break-even point.
PERT	Programme Evaluation and Review Technique — project planning and control tool.	ISRO uses PERT to track the sequence of activities for a satellite launch.
CPM	Critical Path Method — identifies the longest path of activities in a project.	The critical path in building a factory determines the minimum project duration.
MIS	Management Information System — computerised system providing data for decisions.	ERP software giving a store manager real-time sales, inventory, and cost data.
ROI	Return on Investment — profit	ROI = (Net Profit / Total Investment) ×

Term	Simple Meaning	One-line Example
	earned as % of the money invested.	100 = ₹5 lakh / ₹50 lakh × 100 = 10%.

Section 5 Detailed Topic-wise Notes

Topic 1 Meaning of Controlling

A. Simple Meaning

Controlling means comparing what was planned with what actually happened — finding the gap (deviation) — and taking action to fix it so the organisation stays on track toward its goals.

B. CBSE / NCERT Definition

DEFINITION

Controlling is the process of comparing the actual performance with the set standards — finding deviations — and taking corrective action to ensure that organisational activities and results conform to the plans.

Key phrases: 'comparing actual with standards', 'finding deviations', 'corrective action', 'conform to plans'.

C. Features of Controlling

Feature	Meaning	Example
1. Goal-oriented function	Controlling exists only to ensure goals are achieved — it has no purpose without a goal to check against.	Without a target of 90% students passing, there is nothing to 'control' against.
2. Pervasive function	Controlling happens at every level — top, middle, and operational. Every manager controls their area.	CEO controls overall profit. Dept Manager controls department budget. Supervisor controls daily output.
3. Continuous function	Controlling is not one-time — it happens continuously throughout the year.	Monthly sales reviews, weekly production checks, daily attendance monitoring — all continuous control.
4. Dynamic function	As plans change, controls must adapt — controlling is not rigid.	If the market changes and a target becomes unrealistic, the standard itself is revised.
5. Forward-looking function	Good controlling prevents future problems by catching deviations early.	Finding a 10% cost overrun in Month 3 allows corrective action before it becomes 50% by Year-end.

Topic 2 Importance of Controlling

Seven key reasons why controlling is essential. A **common 5-mark question**: 'Explain any five points of importance of controlling.'

1. Accomplishing Organisational Goals

Explanation: Controlling ensures that all activities stay aligned with organisational goals. Without control, individual efforts may diverge from the shared goal.

EXAMPLE

A sales team without controlling might exceed their client entertainment budget by 3× — while achieving only 60% of the sales target. Controlling catches this gap early.

2. Judging Accuracy of Standards

Explanation: Controlling helps evaluate whether the originally set standards were realistic. If actual performance consistently falls short — maybe the standard was too high. Controlling provides evidence to revise standards intelligently.

EXAMPLE

If Smit Sir's standard of 90% students above 80 marks is consistently missed by multiple batches over 2 years, controlling data reveals the standard needs revision.

3. Efficient Use of Resources

Explanation: By comparing actual resource use with planned use, controlling ensures no wastage — of time, money, material, and manpower.

EXAMPLE

A manufacturing firm compares budgeted raw material usage vs actual — finds 15% more material being wasted than planned. Corrective action: stricter quality checks on input.

4. Improving Employee Performance

Explanation: When employees know their performance is being measured and compared against standards, they are more careful, more motivated, and more accountable. Controlling creates performance pressure — positively.

EXAMPLE

Teachers at Smit Sir's institute know their class results are reviewed monthly. This motivates higher preparation and student engagement.

5. Ensures Order and Discipline

Explanation: Controlling ensures that employees follow rules, meet deadlines, and work within defined parameters. It maintains order in the organisation.

EXAMPLE

A company with leave policy controls: if an employee exceeds 15 casual leaves per year, HR automatically processes a salary deduction — maintaining discipline through systematic control.

6. Facilitates Coordination

Explanation: Controlling reveals when different departments are not coordinating well — output of one department doesn't match the input requirements of the next. Controlling enables course correction.

EXAMPLE

If the Marketing Department generates 1,000 leads per month but the Sales team can only handle 400, controlling reveals the mismatch — coordinating to add 2 more salespeople.

7. Psychological Pressure on Employees

Explanation: The knowledge that performance is being measured creates productive psychological pressure — employees stay alert and goal-focused.

EXAMPLE

When students know there is a weekly test (controlling mechanism), they are more likely to study consistently than if there is only one annual exam.

MEMORY TRICK

Memory trick — "A-J-E-I-E-F-P":

A — Accomplishing goals

J — Judging accuracy of standards

E — Efficient use of resources

I — Improving employee performance

E — Ensuring order and discipline
F — Facilitating coordination
P — Psychological pressure on employees

Recall: "All Jobs Exist Inside Every Firm — Properly."

Topic 3 Planning and Controlling — The Twins of Management

Planning and Controlling are called the **'twins of management'** — they are so closely interlinked that neither is complete without the other. This is a **very common 3-5 mark question**.

DEFINITION

Planning is the basis of controlling: Controlling requires standards — and standards come from PLANS. Without a plan, there is nothing to compare actual performance against.

Controlling makes planning meaningful: A plan that is never checked is just a wish. Controlling gives feedback on whether the plan is working — making planning purposeful and actionable.

Together: Planning sets the destination. Controlling checks whether the vehicle is still heading toward that destination — and corrects the course when it drifts.

How Planning and Controlling Interrelate:

Relationship Aspect	Explanation	Example
1. Controlling requires planning	Without a planned standard, there is nothing to control against. The control standard comes from the plan.	Smit Sir cannot check if teaching is on track without first setting the standard: '90% students above 80 marks' from the academic plan.
2. Planning requires controlling	A plan never executed as planned is useless. Controlling ensures the plan is actually being followed.	The academic plan is only meaningful because controlling (monthly tests, result reviews) ensures teachers actually follow it.
3. Both are forward-looking	Both planning and controlling look toward the future — planning creates the future vision; controlling checks whether the present path leads to it.	Planning: 'This year we will achieve ₹100 crore revenue.' Controlling monthly: 'Are we on track for ₹100 crore?'
4. Controlling feeds back into planning	Deviations found through controlling become the basis for improving future plans.	If controlling reveals that 30% of students struggle with Case Studies, NEXT year's plan includes dedicated Case Study sessions.
5. Both are goal-oriented	Planning sets goals. Controlling ensures goals are achieved.	Both exist only because the organisation has GOALS — goals connect planning and controlling.

EXAMPLE

The perfect metaphor: A GPS navigation system in a car.

Planning = The GPS sets the destination and the route (plan).

Controlling = The GPS continuously monitors your actual path. When you deviate — take a wrong turn — it immediately recalculates and tells you how to get back on route.

Without planning: GPS has no destination — nothing to navigate toward.

Without controlling: GPS set a route but never checked if you followed it — you could end up anywhere.

Topic 4 Steps in the Controlling Process — 5 Sequential Steps

These **5 steps** are the **most important part of this chapter** for CBSE. Know all 5 in order, with clear explanation and example for each step.

1	**Setting Performance Standards** *The first step — decide what counts as acceptable performance. Standards can be quantitative (numbers) or qualitative (quality, customer satisfaction).*
2	**Measurement of Actual Performance** *Collect data on what is actually happening. Measure the real output of employees, departments, and processes.*
3	**Comparison of Actual Performance with Standards** *Compare the measured actual performance against the set standard. Calculate the deviation.*
4	**Analysing Deviations** *Understand WHY the deviation occurred. Determine if it is a critical deviation (needs immediate action) or minor (can be monitored).*
5	**Taking Corrective Action** *Fix the problem — change what is being done to bring actual performance in line with standards, or revise the standard if it was unrealistic.*

Step 1: Setting Performance Standards

Explanation: Standards are the yardsticks against which actual performance is measured. They must be set BEFORE work begins — as part of the planning process. Standards can be: Quantitative (sales ₹50 lakh, produce 1,000 units, defects < 2%), Qualitative (customer satisfaction rating > 4.5/5, delivery within 24 hours), Time-based (project delivered by March 31).

EXAMPLE

Smit Sir sets standards: (i) 90% of students score above 80 in Business Studies. (ii) Each teacher completes the syllabus by November 30. (iii) Student satisfaction with teaching: 4.5/5 minimum. These are the standards.

Keywords: standard, benchmark, target, yardstick, quantitative, qualitative, pre-set

Step 2: Measurement of Actual Performance

Explanation: After work begins, the organisation collects data on actual results. Methods of measurement: Personal observation (supervisors watching), statistical reports (data dashboards), written reports (weekly/monthly reports), performance appraisals.

> **EXAMPLE**
>
> After mid-terms: 60% students scored above 80. Syllabus completion: 80% by November 1. Student satisfaction rating: 4.2/5. These are the actual performance figures.

Keywords: measurement, actual, data, observation, reports, survey, count

Step 3: Comparison of Actual Performance with Standards

Explanation: This is the core control step — calculating the deviation. Actual performance is compared against the set standard to find the GAP. This comparison must be done on the SAME basis (same time period, same unit, same measure).

> **EXAMPLE**
>
> Standard vs Actual: (i) 90% - 60% = 30% deviation in student scores. (ii) Syllabus: 100% target vs 80% = 20% incomplete. (iii) Satisfaction: 4.5 - 4.2 = 0.3 below target. These deviations are now visible and measurable.

Keywords: comparison, deviation, gap, actual vs standard, variance, shortfall

Step 4: Analysing Deviations

Explanation: Not all deviations are equally important. This step determines: (a) Which deviations are CRITICAL (large, urgent, need immediate action) and which are MINOR (can be monitored). (b) WHY the deviation occurred — root cause analysis. (c) Which areas are responsible. The principle of 'Management by Exception' says: Focus management attention ONLY on critical deviations. Minor ones can be ignored or monitored.

> **EXAMPLE**
>
> Analysis: (i) 30% gap in student scores = CRITICAL — investigate immediately. Root cause: Students struggle with case study questions. (ii) 0.3 satisfaction gap = MINOR — monitor for next month. (iii) Syllabus: 20% gap = significant — need to accelerate pace.

Keywords: analyse, critical deviation, minor deviation, management by exception, root cause, why

Step 5: Taking Corrective Action

Explanation: Based on the analysis, management takes action to fix the problem. Corrective action can be: (a) Fixing the performance (changing how work is done), (b) Revising the standard (if it was unrealistic), or (c) Both.

> **EXAMPLE**

Corrective action: (i) Add 3 extra case study practice sessions per week. (ii) Create a 'Case Study Formula' guide for students. (iii) Accelerate syllabus pace for next 2 weeks. (iv) Schedule a re-test in January to measure improvement. These actions bring actual performance back toward the standard.

Keywords: corrective action, fix, revise standard, adjust, course correct

Key Concepts Within the Controlling Process

Concept	Meaning	Example
Critical deviation	A deviation large enough to demand IMMEDIATE management attention and corrective action.	30% shortfall in student scores — too large to ignore. Immediate action required.
Minor deviation	A small deviation — within acceptable range — that can be monitored without immediate action.	Syllabus completion 97% vs 100% target — 3% gap is minor, no urgent action needed.
Management by Exception	Principle: Managers should focus their limited time and attention ONLY on significant (critical) deviations — not on every small variance.	Factory manager reviews only the machines with >10% efficiency drop — not every machine every day.
Revision of standards	If actual performance consistently cannot meet a standard despite best efforts, the standard itself may need revision.	If 90% above 80 marks is never achieved across 5 batches over 3 years — the standard may be unrealistic and needs adjustment.

MEMORY TRICK

Memory trick — 5 steps — "S-M-C-A-T":

S — Setting performance standards

M — Measurement of actual performance

C — Comparison of actual with standards

A — Analysing deviations

T — Taking corrective action

Recall: "Students Must Compare Actual Tests."

Topic 5 Techniques of Managerial Control

There are two categories of control techniques: **Traditional** and **Modern**. These techniques are the **TOOLS** used in Steps 1 and 2 of the controlling process (setting standards and measuring actual performance).

A. TRADITIONAL TECHNIQUES OF CONTROL

T1 Personal Observation

DEFINITION

The manager or supervisor physically visits the workplace and personally observes employees at work — watching how they perform, checking quality, and giving immediate feedback.

Feature	Detail
Method	Direct, first-hand observation of work by the manager.
Best for	Operational/supervisory level — factory floor, field operations, classroom.
Advantage	Provides real, unfiltered information. Allows immediate guidance and correction.
Disadvantage	Time-consuming. Cannot cover large organisations. May affect employee behaviour when being watched (Hawthorne effect).

EXAMPLE

Example: Smit Sir himself sitting in on teacher classes once a week to observe teaching quality, student engagement, and coverage of concepts. Personal observation of teaching = first-hand, real-time control information.

T2 Statistical Reports

DEFINITION

Statistical reports involve analysing and presenting data through graphs, charts, tables, averages, and statistical measures — to make performance trends visible and

easy to compare with standards.

Feature	Detail
Method	Collecting data, presenting it visually (bar charts, pie charts, trend lines, averages).
Best for	Tracking trends over time — sales, production volume, cost trends, quality rates.
Advantage	Easy to spot trends. Visual and clear. Can cover large amounts of data at once.
Disadvantage	Statistical reports can be manipulated. Non-quantifiable aspects (morale, creativity) cannot be captured.

EXAMPLE

Example: A monthly report showing: batch-wise results, topic-wise error analysis, teacher-wise class performance, and a trend line of average marks over 6 months. Smit Sir uses this dashboard to spot declining performance in specific areas before they become critical.

T3 Break-even Analysis

DEFINITION

Break-even analysis is a technique that determines the level of sales or production at which total revenue equals total cost — the 'break-even point' (BEP) — where there is neither profit nor loss.

Feature	Detail
Formula	Break-Even Point (units) = Fixed Costs / (Selling Price per unit − Variable Cost per unit)
Purpose	Tells the manager the minimum sales needed to cover all costs — a control standard for viability.
Control use	If actual sales are below BEP — the business is running at a loss. Action must be taken.
Advantage	Simple, powerful planning and control tool. Helps in pricing decisions.
Disadvantage	Assumes all units are sold at the same price; ignores volume discounts and demand changes.

EXAMPLE

Feature	Detail

Example: Smit Sir's new branch: Fixed costs = ₹1,50,000/month (rent, salaries). Fee per student = ₹3,000. Variable cost per student = ₹500. Contribution = ₹2,500 per student.

BEP = ₹1,50,000 / ₹2,500 = **60 students** — the branch must enroll at least 60 students to break even.

If actual enrollment is 45 students — below BEP — Smit Sir must take corrective action: increase marketing, add a new course, or reduce fixed costs.

T4 Budgetary Control

DEFINITION

Budgetary control is the process of comparing actual financial performance (revenue, expenditure, profit) with the budgeted (planned) figures — identifying variances — and taking corrective action.

Feature	Detail
What is a budget?	A detailed financial plan — expected revenue, costs, and profit for a specific period.
How it controls	Actual figures are compared with budgeted figures. Variance = actual − budget.
Types of variance	Favourable variance: actual better than budget. Unfavourable variance: actual worse than budget.
Control action	Large unfavourable variances trigger investigation and corrective action.
Advantage	Covers entire organisation. Quantifiable and objective. Historical record for future planning.
Disadvantage	Budgets can become rigid — may resist adapting to changed conditions. Risk of 'budget gaming' — departments spend all budget to avoid cuts next year.

EXAMPLE

Example: Smit Sir budgets: Marketing cost = ₹50,000/month. Actual spent = ₹75,000. Variance = ₹25,000 unfavourable.

He also budgets: Fee revenue = ₹2,00,000/month. Actual = ₹2,30,000. Variance = ₹30,000 favourable.

Budgetary control: Marketing overspent (investigate why). Revenue exceeded (congratulate marketing team). Net position: still profitable despite overspend.

B. MODERN TECHNIQUES OF CONTROL

M1 Return on Investment (ROI)

DEFINITION

ROI measures the profitability of an investment — it expresses the net profit earned as a percentage of the total capital invested.

Formula: ROI = (Net Profit / Total Investment) × 100

Feature	Detail
Purpose	Measures how efficiently the organisation is using its capital to generate profit.
Control use	Compare ROI across departments, products, or years to identify which investments are performing and which need attention.
Advantage	Simple, universal metric. Allows comparison across businesses, departments, and investments.
Limitation	Does not consider time value of money. Can be manipulated by accounting methods.

EXAMPLE

Example: Branch A: Net profit ₹5 lakh on investment of ₹50 lakh → ROI = 10%. Branch B: Net profit ₹4 lakh on investment of ₹20 lakh → ROI = 20%. Despite Branch B earning less absolute profit, its ROI is double — Branch B is using capital more efficiently.

Smit Sir uses ROI to decide: should he open Branch C in Ahmedabad or invest in an online platform? ROI comparison guides the decision.

M2 Ratio Analysis

DEFINITION

Ratio analysis involves calculating financial ratios from balance sheet and profit & loss account data to analyse the financial health, efficiency, and profitability of the business.

Type of Ratio	What it measures	Example
Liquidity ratios	Ability to pay short-term debts (Current ratio, Quick ratio).	Current ratio = Current assets / Current liabilities. Ideal = 2:1.
Solvency ratios	Ability to meet long-term obligations (Debt-equity ratio).	Debt-Equity ratio = Total Debt / Equity. Lower = less risky.
Profitability ratios	How well the business converts sales to profit (Net profit margin, ROE).	Net profit margin = Net profit / Sales × 100.
Activity/Efficiency ratios	How efficiently resources are used (Inventory turnover, Debtor turnover).	Inventory turnover = Cost of goods sold / Average inventory.

EXAMPLE

Example: A ratio analysis reveals Smit Sir's coaching institute: Current Ratio = 3:1 (very liquid — good). Profit margin = 25% (healthy). But Debtor turnover is low — students are paying fees late. Control action: tighten fee collection policy.

M3 Responsibility Accounting

DEFINITION

Responsibility accounting is a system where each manager is held responsible and accountable for the revenue, costs, or profits of their specific department or division — their 'responsibility centre'.

Type of Centre	Responsibility	Example
Cost centre	Manager responsible for controlling costs in their area.	Production manager responsible for keeping per-unit manufacturing cost within ₹500.
Revenue centre	Manager responsible for achieving revenue targets.	Sales manager responsible for ₹2 crore monthly sales target.
Profit centre	Manager responsible for both revenue and costs — hence profit.	Branch manager responsible for the branch's profitability.
Investment centre	Manager responsible for profit AND the assets/investments used.	Division head responsible for ROI of their entire division.

EXAMPLE

Example: Smit Sir makes each teacher 'responsible' for their batch results (revenue centre for batch enrollment + cost centre for teaching materials). Each teacher's performance is

Type of Centre	Responsibility	Example
evaluated against their specific batch's enrollment and results — clear, accountable control.		

M4 Management Audit

DEFINITION

Management audit is a comprehensive and systematic review of the management practices, organisational structure, and managerial efficiency of an enterprise — conducted by an independent body or expert team.

Feature	Detail
Purpose	To evaluate the overall effectiveness of management — are the right decisions being made?
Who conducts	Independent experts, internal audit team, or external consultants.
What it covers	Goals, policies, structure, procedures, use of resources, decision quality, communication.
Advantage	Comprehensive overview. Independent and objective. Identifies systemic weaknesses.
Limitation	Expensive. Time-consuming. Disruptive if not well-planned.

EXAMPLE

Example: A management audit of Smit Sir's institute evaluates: Are teachers following the curriculum? Is the fee structure competitive? Are administrative costs justified? Are student grievances addressed within 48 hours? This comprehensive review reveals gaps that individual metrics miss.

M5 PERT and CPM

DEFINITION

PERT (Programme Evaluation and Review Technique) and **CPM (Critical Path Method)** are network analysis techniques used to plan, schedule, and control complex projects with many interrelated activities.

PERT: Used when activity times are **uncertain** (three time estimates: optimistic, most likely, pessimistic).

CPM: Used when activity times are **known** — identifies the critical path (longest sequence of activities) that determines minimum project duration.

Feature	PERT	CPM
Time estimates	Probabilistic (uncertain — 3 estimates)	Deterministic (known — 1 estimate)
Focus	Time management in uncertain projects	Cost + time management in well-defined projects
Best for	Research, new product development, complex R&D	Construction, manufacturing, well-defined projects
Critical path	Identifies critical path	Specifically designed around critical path analysis

EXAMPLE

Example: ISRO uses PERT to manage the Chandrayaan mission — thousands of activities (spacecraft design, testing, launch window, trajectory calculation) with uncertain durations. The PERT chart ensures all activities are tracked and the mission doesn't get delayed. A construction company uses CPM for building a factory — each task (foundation, structure, wiring, plumbing) has known duration — CPM finds the critical path (21 weeks) and ensures on-time completion.

M6 Management Information System (MIS)

DEFINITION

Management Information System (MIS) is a computerised system that collects, stores, processes, and presents information to managers at all levels — enabling better, faster, and more accurate decision-making and control.

Feature	Detail
Input	Raw data from operations — sales, inventory, costs, customer feedback.
Processing	Data is organised, analysed, and converted into meaningful information.
Output	Reports, dashboards, alerts — presented to managers for decision-making.
Control use	Provides real-time data on performance vs standards — enables rapid deviation detection.

Feature	Detail
Advantage	Fast. Accurate. Covers entire organisation simultaneously. Reduces human error.
Limitation	Expensive to implement. Requires data accuracy — 'garbage in, garbage out'. System failures can paralyse operations.

EXAMPLE

Example: Amazon India's MIS tracks in real time: orders, delivery status, inventory levels, ratings, complaints, return rates, and revenue by category. Any deviation triggers an automatic alert for immediate corrective action — MIS-enabled controlling at massive scale.

All Control Techniques — Quick Reference

Category	Technique	What it does	Best used for
Traditional	1. Personal Observation	Manager personally watches work at site.	Operational level — factory, classroom.
Traditional	2. Statistical Reports	Data presented as charts, averages, trends.	Spotting trends across large datasets.
Traditional	3. Break-even Analysis	Finds minimum sales to cover all costs.	Pricing and viability decisions.
Traditional	4. Budgetary Control	Compares actual financial results with budget.	Financial performance control.
Modern	5. ROI	Measures profit as % of investment.	Comparing efficiency across investments.
Modern	6. Ratio Analysis	Financial ratios assess health and efficiency.	Overall financial health diagnosis.
Modern	7. Responsibility Accounting	Holds each manager responsible for their area.	Divisional / departmental accountability.
Modern	8. Management Audit	Comprehensive review of management practices.	Identifying systemic weaknesses.
Modern	9. PERT and CPM	Network analysis for project planning and control.	Complex, multi-activity projects.
Modern	10. MIS	Computer system providing real-time performance data.	Large organisations needing rapid control.

MEMORY TRICK

Traditional techniques (P-S-B-B): Personal observation, Statistical reports, Break-even analysis, Budgetary control.

Modern techniques (R-R-R-M-P-M): ROI, Ratio analysis, Responsibility accounting, Management audit, PERT/CPM, MIS.

Recall traditional: "People Share Budgets Boldly."
Recall modern: "Really Reliable Results Matter — Plan More Intelligently."

Section 7 Critical Difference Tables

7.1 Planning vs Controlling

This **difference appears as a 4-5 mark question** in CBSE boards.

Basis	Planning	Controlling
Meaning	Deciding in advance what is to be done and how.	Ensuring that activities conform to plans and taking corrective action if they don't.
Nature	Future-oriented — looks ahead.	Both past (compares actual results) and future (takes corrective action for future).
First step	Precedes all other functions — it is the first function.	Follows all other functions — it is the last function.
Basis	Plans are based on forecasts and objectives.	Controls are based on set plans and standards.
Focus	Formulation of goals and strategies.	Achievement of goals and performance measurement.
Relationship	Planning provides the standard for controlling.	Controlling checks whether planning is effective.
Type	Primarily a thinking function.	Primarily a checking and corrective action function.

7.2 Traditional vs Modern Techniques of Control

Basis	Traditional Techniques	Modern Techniques
Nature	Simple, mostly non-quantitative, direct.	Sophisticated, quantitative, often technology-aided.
Data requirement	Minimal — observation, basic data.	Requires detailed financial data or computer systems.
Coverage	Usually limited to specific areas or levels.	Can cover entire organisation simultaneously (especially MIS).
Speed	Slower — manual collection and analysis.	Faster — computerised systems provide real-time data.
Examples	Personal observation, budgetary control.	MIS, PERT/CPM, ROI, Ratio analysis.

Section 9 All Memory Tricks at One Place

Topic	Trick	Expansion
5 Steps of Controlling	S-M-C-A-T	Setting standards, Measuring actual, Comparing, Analysing deviations, Taking corrective action
7 Importance	A-J-E-I-E-F-P	Accomplishing goals, Judging standards, Efficient resources, Improving performance, Ensuring discipline, Facilitating coordination, Psychological pressure
Traditional techniques	P-S-B-B	Personal observation, Statistical reports, Break-even analysis, Budgetary control
Modern techniques	R-R-R-M-P-M	ROI, Ratio analysis, Responsibility accounting, Management audit, PERT/CPM, MIS
Planning-Controlling relationship	Twins of management	Neither can exist without the other — planning sets standards, controlling checks them
Management by Exception	Focus on CRITICAL only	Only deviations large enough to demand action get management attention

Section 10 Case Study Practice

Case Study 1 — Easy (Identify the Controlling Step)

CASE STUDY

Case: Map each action below to the correct step in the controlling process:

(a) Smit Sir decides: 'Every teacher must complete the full syllabus by November 30 and achieve minimum 4.5/5 student satisfaction.'

(b) Smit Sir collects November exam results, attendance data, and student feedback forms.

(c) He compares: Syllabus completion actual = 85% vs target 100%. Satisfaction: 4.3/5 vs 4.5/5 target.

(d) He identifies that the shortfall in syllabus completion is in the new 'Finance' teacher's batch. But the 0.2 satisfaction gap is a minor deviation — all experienced teachers are at 4.7-4.8.

(e) He arranges an extra 3 classes per week for the Finance batch and pairs the new teacher with a senior mentor.

Answers:

(a) → **Step 1: Setting performance standards** — targets for syllabus and satisfaction are being set.

(b) → **Step 2: Measurement of actual performance** — exam results and feedback data are collected.

(c) → **Step 3: Comparison of actual with standards** — actual vs target comparison = deviation identified.

(d) → **Step 4: Analysing deviations** — critical deviation (Finance batch shortfall) vs minor deviation (satisfaction gap) separated.

(e) → **Step 5: Taking corrective action** — extra classes and mentor are the corrective actions.

Case Study 2 — Moderate (Identify the Control Technique)

CASE STUDY

Case: Identify the technique of managerial control used in each situation:

(a) The HR Director of a manufacturing company walks the factory floor every morning at 8 AM to check if workers have arrived on time, machines are clean, and safety protocols are followed.

(b) A construction company creates a detailed network chart showing all activities needed to complete a building — from foundation to finishing — identifying which activities cannot be delayed without delaying the whole project.

(c) Reliance Retail's system automatically alerts the store manager when any product's actual sales fall more than 20% below the forecasted sales for that week.

(d) A food company calculates: if it sells 10,000 packets/month, it covers all fixed and variable costs exactly. Anything above that is profit. It uses this as a benchmark for viability.

(e) An independent team reviews the entire management of 'Sunrise Ltd.' — including decision-making quality, communication effectiveness, resource utilisation, and goal achievement — and provides a comprehensive improvement report.

Answers:

(a) → **Personal Observation** — direct, first-hand watch by the HR Director.

(b) → **CPM (Critical Path Method)** — network of activities with critical path identification for construction project.

(c) → **MIS (Management Information System)** — computerised real-time alert system comparing actual vs forecast.

(d) → **Break-even Analysis** — finding the minimum sales (10,000 packets) to break even.

(e) → **Management Audit** — comprehensive, independent review of all management practices.

Case Study 3 — Difficult / HOTS (Controlling Process + Planning Link)

CASE STUDY

Case: 'FreshBite Ltd.' manufactures healthy snacks. They set annual targets: Sales = ₹5 crore, Production = 10 lakh units, Defects < 1%, Marketing budget = ₹30 lakh. After 6 months, the data shows: Sales = ₹2 crore (40% of annual target), Production = 4.2 lakh units, Defects = 3.5%, Marketing spent = ₹28 lakh (93% of annual budget already used).

Q1: Apply the 5-step controlling process to this case.

Q2: Which deviations are critical and which are minor? Apply the principle of Management by Exception.

Q3: How does this case illustrate the relationship between planning and controlling?

Answer Q1 — Controlling process applied:

Step 1 (Standards): Sales ₹5 cr, Production 10 lakh units, Defects <1%, Marketing ₹30 lakh.

Step 2 (Measurement): Sales ₹2 cr, Production 4.2 lakh, Defects 3.5%, Marketing ₹28 lakh in 6 months.

Step 3 (Comparison): Sales at 40% (expected ~50%) = -10% shortfall. Production at 42% (expected ~50%) = -8%. Defects 3.5% vs <1% = +2.5%. Marketing 93% spent in 50% of time = massive overspend.

Step 4 (Analysis): Critical: Defects (3.5% vs <1%) = 2.5% above standard — quality crisis. Marketing overspend = 43% over budget in half the year = critical cash drain. Moderate: Sales and Production shortfall — investigate production capacity constraint.

Step 5 (Corrective action): (i) Immediate quality audit — find root cause of 3.5% defect rate. (ii) Freeze remaining marketing spend until review. (iii) Add 1 production shift to address units shortfall. (iv) Revise marketing budget quarterly.

Answer Q2 — Management by Exception:

Critical deviations (require immediate action): Defect rate (3.5% vs <1% = 250% above standard) and Marketing overspend (93% used in 50% of time = critical). Both need immediate management attention.

Minor deviations (monitor): Sales shortfall (-10% vs expected) and Production shortfall (-8%) are moderate — within range, may self-correct with increased production.

Answer Q3 — Planning-Controlling link:

This case perfectly illustrates the twins relationship: The **annual targets (plan)** — ₹5 cr sales, <1% defects, ₹30 lakh marketing — provide the **standards for controlling**. Without these planned standards, the 6-month data is just random numbers with no meaning. Controlling (comparing actual vs plan) reveals that the **plan was overoptimistic** on marketing budget, providing feedback for REVISION of next year's plan — completing the planning-controlling cycle.

Section 11 Exam-Oriented Questions & Answers

A. One-mark Questions

Q1. Define controlling.

Ans. Controlling is the process of comparing actual performance with set standards, identifying deviations, and taking corrective action to ensure activities conform to plans.

Q2. Name the five steps in the controlling process.

Ans. Setting standards, Measuring actual performance, Comparing with standards, Analysing deviations, Taking corrective action (S-M-C-A-T).

Q3. What is 'deviation' in controlling?

Ans. Deviation is the difference between actual performance and the set standard.

Q4. What is 'Management by Exception'?

Ans. Management by Exception means managers focus their attention only on significant (critical) deviations — minor deviations are allowed or monitored without immediate action.

Q5. Name any two traditional techniques of controlling.

Ans. Personal Observation and Budgetary Control (or any two from P-S-B-B).

Q6. Name any two modern techniques of controlling.

Ans. ROI and MIS (or any two from R-R-R-M-P-M).

Q7. Why are planning and controlling called 'twins of management'?

Ans. Because they are inseparable — planning sets standards and controlling uses those standards to check performance. Neither is useful without the other.

Q8. What is 'break-even point'?

Ans. Break-even point is the level of sales or production at which total revenue equals total cost — no profit, no loss.

B. Three-mark Questions

Q1. Explain the relationship between planning and controlling.

Ans. Planning and controlling are called the 'twins of management' — they are inseparable.

Ans. (i) Controlling requires planning: Without planned standards, there is nothing to compare actual performance against. Planning provides the benchmarks that controlling uses.

Ans. (ii) Planning requires controlling: A plan without control is merely a wish. Controlling ensures that the plan is actually being executed and provides feedback to improve future plans. Together, they form a continuous cycle: Plan → Execute → Control → Revise Plan.

Q2. Explain the steps in the controlling process.

Ans. The controlling process has 5 steps (S-M-C-A-T):

Ans. (i) Setting performance standards: Deciding the benchmarks — targets, goals, quality standards — against which actual performance will be measured.

Ans. (ii) Measuring actual performance: Collecting real data on what is happening — through observation, reports, and appraisals.

Ans. (iii) Comparing actual with standards: Finding the deviation — actual minus standard = gap.

Ans. (iv) Analysing deviations: Identifying critical vs minor deviations and finding root causes.

Ans. (v) Taking corrective action: Fixing the gap — changing how work is done or revising the standard if it was unrealistic.

Q3. Explain any three traditional techniques of managerial control.

Ans. (i) Personal Observation: The manager visits the workplace and directly observes employee performance — provides real, first-hand information. Example: Factory supervisor walking the floor daily.

Ans. (ii) Statistical Reports: Data on performance is presented through graphs, charts, and tables — making trends visible and comparable. Example: Monthly sales trend chart.

Ans. (iii) Budgetary Control: Actual financial performance (revenue, costs) is compared with the budget — unfavourable variances trigger investigation. Example: Sales ₹45 lakh vs budget ₹50 lakh → ₹5 lakh shortfall to investigate.

C. Five & Six-mark Questions

Q1. Explain the importance of controlling in an organisation.

Ans. Controlling is important for the following reasons:

• **Accomplishing goals:** Controlling ensures all activities stay aligned with organisational goals — deviations are caught and fixed.

• **Judging accuracy of standards:** Controlling reveals whether set standards were realistic — provides evidence to revise them.

• **Efficient use of resources:** By comparing actual vs planned resource use, controlling eliminates wastage of money, time, and material.

• **Improving employee performance:** Knowing performance is measured makes employees more careful and accountable.

• **Ensuring order and discipline:** Controlling maintains adherence to rules, deadlines, and procedures.

• **Facilitating coordination:** Reveals inter-departmental mismatches — enabling timely coordination.

• **Psychological pressure:** The knowledge of being measured creates productive focus among employees.

Q2. Distinguish between planning and controlling on any five bases.

Ans.

• **Nature:** Planning is forward-looking only. Controlling is both backward (compares actual results) and forward (takes corrective action for future).

• **Order:** Planning is the FIRST management function. Controlling is the LAST function.

• **Focus:** Planning focuses on deciding what to do. Controlling focuses on checking whether it is being done.

• **Basis:** Plans are based on forecasts and objectives. Controls are based on set plans.

• **Relationship:** Planning provides standards for controlling. Controlling validates and improves planning.

D. MCQs

Q1. The first step in the controlling process is —

(a) Measuring actual performance

(b) Taking corrective action

(c) Setting performance standards

(d) Analysing deviations

Ans. (c) Setting performance standards — you must set the benchmark before you can measure or compare.

Q2. 'Management by Exception' means —

(a) Managing only exceptional employees

(b) Focusing on all deviations equally

(c) Ignoring all deviations

(d) Focusing only on critical deviations

Ans. (d) Focusing only on critical deviations — minor deviations are allowed without management intervention.

Q3. Planning and Controlling are called —

(a) Functions of management

(b) Twins of management

(c) Principles of management

(d) Tools of management

Ans. (b) Twins of management — they are inseparable and interdependent.

Q4. Break-even analysis determines —

(a) Maximum profit level

(b) Minimum investment needed

(c) Level where revenue = cost, no profit or loss

(d) Maximum revenue possible

Ans. (c) The level at which total revenue equals total cost — the break-even point.

Q5. PERT is used when —

(a) Activity times are known with certainty

(b) Project has only 2 activities

(c) Activity times are uncertain

(d) Only financial control is needed

Ans. (c) Activity times are uncertain — PERT uses three time estimates (optimistic, most likely, pessimistic).

E. Assertion-Reason

Q. Assertion (A): Planning and controlling are inseparable functions of management.

Reason (R): Planning provides standards that controlling uses to measure performance, and controlling provides feedback that improves future plans.

Options: (a) A and R true, R correctly explains A. (b) Both true, R does NOT explain. (c) A true, R false. (d) A false, R true.

Ans. (a) — Both correct and R precisely explains why A is true. The mutual dependence between planning and controlling is exactly what makes them inseparable — each depends on the other to be meaningful.

Section 12 Board Exam Answer Writing Guide

Case-study trigger table — Identify concept quickly

If case says...	Concept
Manager sets target/benchmark before work begins	Step 1: Setting performance standards
Data on actual results collected — exam, reports, observation	Step 2: Measuring actual performance
Actual vs target — gap/variance calculated	Step 3: Comparison of actual with standards
Identifying which deviations need action and why they occurred	Step 4: Analysing deviations
Extra sessions added, process changed, standard revised	Step 5: Taking corrective action
Manager focuses only on large deviations, ignores small ones	Management by Exception / Critical deviation
Manager personally walks factory/office to observe	Personal Observation (Traditional technique)
Monthly performance data shown in chart/graph form	Statistical Reports (Traditional)
Minimum sales/output to cover all costs	Break-even Analysis (Traditional)
Actual vs budgeted revenue/cost comparison	Budgetary Control (Traditional)
Net profit as % of investment	Return on Investment — ROI (Modern)
Financial ratios — current ratio, profit margin	Ratio Analysis (Modern)
Each manager responsible for their cost/revenue/profit	Responsibility Accounting (Modern)
Independent team reviews entire management practices	Management Audit (Modern)
Network chart showing all project activities, critical path	PERT/CPM (Modern)
Computer system providing real-time performance data	MIS — Management Information System (Modern)
Planning sets target; actual checked against it — cycle	Planning and Controlling Relationship

EXAM TIP

For 5-step controlling process question: Write all 5 steps in ORDER (S-M-C-A-T). Name each step clearly. Give a consistent example across all 5 steps (same company, same scenario) — it shows clear understanding.

For importance of controlling: Write 5-6 points with brief explanation + example for each. Never just list names — always explain WHY that point makes controlling important.

For techniques question: Divide into Traditional (P-S-B-B) and Modern (R-R-R-M-P-M) categories first. Define each technique + one Indian example. This structure shows organised thinking.

Section 13 Common Doubts Students Ask

Doubt 1: Sir, is controlling the last function of management?

Smit Sir's answer: Yes — in the sequence: Planning → Organising → Staffing → Directing → Controlling. Controlling is the last. But it's also the link back to planning — its output (deviations, corrective action insights) feeds INTO the next planning cycle. So the management cycle is circular, not just linear.

Doubt 2: Sir, what is the difference between deviation and variance?

Smit Sir's answer: In CBSE context, they are used interchangeably. Deviation = difference between actual performance and standard. Variance = same concept in financial/budgetary control. Unfavourable variance = actual worse than planned. Favourable variance = actual better than planned.

Doubt 3: Sir, can the standard itself be wrong?

Smit Sir's answer: Yes — and Step 4 (analysing deviations) must consider this. If actual performance consistently falls short of the standard despite best efforts, the standard may be unrealistic. Corrective action in Step 5 can then be: revise the standard (not just the performance).

Doubt 4: Sir, is PERT or CPM better?

Smit Sir's answer: Neither is universally better — they are complementary. PERT for uncertain-duration projects (R&D, new product development). CPM for known-duration, cost-focused projects (construction, manufacturing). In practice, both are often used together on complex projects.

Doubt 5: Sir, what is the difference between MIS and statistical reports?

Smit Sir's answer: Statistical Reports = manually produced charts and tables from collected data. MIS = a computerised, automated system that collects, processes, and presents data in real time. MIS is faster, covers more data, and works continuously — statistical reports are prepared periodically (weekly, monthly) by humans.

Doubt 6: Sir, why does controlling need planning? Can we control without a plan?

Smit Sir's answer: Without a plan, there is no standard to compare against. If you have no target (no plan), actual performance has no meaning — is 60 marks good or bad? You don't know without a standard. Controlling without planning is like judging a race without knowing the finish line.

Doubt 7: Sir, is budgetary control only about money?

Smit Sir's answer: Primarily yes — it compares actual financial figures (revenue, costs, profit) against budgeted figures. But budgets can also be non-financial: labour hours budget, units of production budget, material consumption budget. Budgetary control covers any budgeted resource.

Doubt 8: Sir, what is the difference between Management Audit and Financial Audit?

Smit Sir's answer: Financial Audit = independent verification of financial statements (done by Chartered Accountants for legal compliance). Management Audit = comprehensive review of management practices, decision quality, and organisational effectiveness (done by management consultants or internal audit teams for improvement, not compliance).

Doubt 9: Sir, why is controlling called a 'forward-looking' function when it compares past data?

Smit Sir's answer: Controlling looks at PAST performance (actual vs standard) — but the PURPOSE is to improve FUTURE performance. The corrective action taken addresses a future challenge. So controlling has both a backward-looking component (measurement and comparison) and a forward-looking outcome (corrective action and improved plans).

Doubt 10: Sir, does controlling only happen at top management level?

Smit Sir's answer: No — controlling is PERVASIVE — it happens at all levels. Top management controls strategic goals (profit, market share). Middle management controls departmental targets. Supervisors/foremen control daily output, attendance, and quality. Every manager controls their area of responsibility.

Section 14 Common Mistakes to Avoid

Mistake 1: Writing controlling steps out of order

What students do wrong: Writing 'Taking corrective action' before 'Analysing deviations'.

Why it is wrong: The 5 steps MUST follow S-M-C-A-T. You cannot take corrective action before you analyse why the deviation occurred.

Correct way: Memorise S-M-C-A-T. NEVER skip Step 4 (Analyse). You cannot jump from comparison directly to corrective action.

Mistake 2: Saying planning comes after controlling

What students do wrong: 'Controlling comes first, then planning.'

Why it is wrong: WRONG. Planning is the FIRST function. Controlling is the LAST. Controlling's outputs feed back into the NEXT planning cycle — but within one cycle, Planning comes first.

Correct way: Sequence: Planning → Organising → Staffing → Directing → Controlling. Controlling then informs the next round of Planning.

Mistake 3: Treating all deviations as equally important

What students do wrong: 'The manager must investigate every deviation — large and small.'

Why it is wrong: WRONG. Management by Exception says: focus ONLY on critical/significant deviations. Investigating every minor deviation wastes management time.

Correct way: Critical deviation = requires immediate action. Minor deviation = monitor or accept. Management by Exception = key principle.

Mistake 4: Confusing PERT and CPM

What students do wrong: 'PERT and CPM are the same thing — both are project management tools.'

Why it is wrong: While both are network techniques, PERT is for UNCERTAIN duration (three time estimates) and CPM is for KNOWN duration (one time estimate, cost-focused).

Correct way: PERT = Probabilistic (uncertain times, R&D). CPM = Deterministic (known times, construction). Both find the critical path but handle time differently.

Mistake 5: Saying break-even analysis gives maximum profit

What students do wrong: 'Break-even analysis shows the maximum profit a company can earn.'

Why it is wrong: Break-even analysis shows the MINIMUM sales to cover costs (zero profit, zero loss). Anything ABOVE the break-even point generates profit.

Correct way: BEP = no profit, no loss. Above BEP = profit. Below BEP = loss. BEP is about MINIMUM VIABILITY, not maximum profit.

Mistake 6: Not including examples in controlling step answers

What students do wrong: Writing only definitions for each step without a practical example.

Why it is wrong: CBSE examiners look for application of concepts. An answer without examples scores below full marks.

Correct way: For EVERY step, use a consistent, relatable example (coaching class, factory, supermarket) applied across all 5 steps. Continuity of example shows conceptual clarity.

Section 15 Keywords Bank

Topic	Must-use Keywords
Controlling — definition	comparing actual with standards, deviations, corrective action, conform to plans, goal-oriented
Importance	accomplishing goals, efficient resources, employee performance, discipline, coordination, psychological pressure
Planning-Controlling relationship	twins of management, inseparable, standards from plans, feedback to planning, forward-looking, circular
5 Steps (S-M-C-A-T)	setting standards, measurement, comparison, deviation, critical deviation, minor deviation, corrective action
Management by Exception	critical deviation, significant, focus attention, ignore minor, management time
Traditional techniques	personal observation, statistical reports, break-even analysis, budgetary control, variance, favourable, unfavourable
Modern techniques	ROI, ratio analysis, responsibility accounting, management audit, PERT, CPM, MIS, real-time, network, critical path

Section 16 Quick Revision Notes

QUICK REVISION

Controlling: Comparing actual with planned standards → finding deviations → taking corrective action. Features: goal-oriented, pervasive, continuous, dynamic, forward-looking.

7 Importance (A-J-E-I-E-F-P): Accomplishing goals, Judging standards, Efficient resources, Improving performance, Ensuring discipline, Facilitating coordination, Psychological pressure.

Planning and Controlling = Twins: Planning sets standards. Controlling checks them. Control feedback improves plans. Neither exists without the other.

5 Steps (S-M-C-A-T):

1. Setting standards → 2. Measuring actual → 3. Comparing → 4. Analysing deviations (critical vs minor) → 5. Taking corrective action.

Management by Exception: Focus only on CRITICAL deviations. Ignore minor ones.

Traditional techniques (P-S-B-B): Personal Observation, Statistical Reports, Break-even Analysis, Budgetary Control.

Modern techniques (R-R-R-M-P-M): ROI, Ratio Analysis, Responsibility Accounting, Management Audit, PERT/CPM, MIS.

BEP Formula: Fixed Costs / (Selling price per unit - Variable cost per unit) = Break-even point in units.

ROI Formula: (Net Profit / Total Investment) × 100 = Return on Investment %.

Section 17 One-Page Last-Minute Revision Sheet

Read this on exam day. Nothing else.

Chapter in one paragraph

Controlling = comparing actual with standards → finding deviations → corrective action. It is the last function of management and the **twin of planning** (planning sets standards; controlling checks them). The **5-step process (S-M-C-A-T)**: Setting standards → Measuring actual → Comparing → Analysing deviations (**critical vs minor**, **Management by Exception**) → Taking corrective action. Techniques: **Traditional (P-S-B-B)**: Personal Observation, Statistical Reports, Break-even Analysis, Budgetary Control. **Modern (R-R-R-M-P-M)**: ROI, Ratio Analysis, Responsibility Accounting, Management Audit, PERT/CPM, MIS.

5 must-remember lines

11. 5 steps: S-M-C-A-T. Setting standards FIRST. Corrective action LAST.
12. Planning and Controlling = Twins. Planning provides standards. Controlling checks and improves planning.
13. Management by Exception = focus ONLY on critical deviations. Minor deviations can be monitored.
14. Break-even = Fixed cost / Contribution per unit. BEP = zero profit, zero loss.
15. PERT = uncertain duration (probabilistic). CPM = known duration (deterministic). Both = network techniques.

Quick trigger table

If case says...	Answer
Manager sets target/benchmark before starting work	Step 1: Setting performance standards
Collecting actual data — exam scores, sales figures, reports	Step 2: Measuring actual performance
Actual vs standard — gap calculated	Step 3: Comparison of actual with standards
Finding why deviation occurred, critical vs minor	Step 4: Analysing deviations
Adding sessions, changing process, revising target	Step 5: Taking corrective action
Manager investigates only large gaps, ignores small	Management by Exception
Manager walks factory/office to observe directly	Personal Observation
Sales trend chart, monthly data dashboard	Statistical Reports

If case says...	Answer
Minimum units/sales to break even	Break-even Analysis
Actual vs budgeted revenue/costs compared	Budgetary Control
Profit as % of investment	ROI
Current ratio, debt-equity, profit margin	Ratio Analysis
Each manager responsible for their centre's results	Responsibility Accounting
Independent full review of management practices	Management Audit
Network chart, critical path, project timeline	PERT / CPM
Computer system, real-time alerts, automated reporting	MIS

Section 18 How Smit Sir Can Teach This Chapter

1. Opening hook — CBSE exam as a control mechanism

Tell students: 'The CBSE board exam itself is a controlling mechanism. The standard = the CBSE prescribed syllabus. The measurement = your marks. The comparison = marks vs the grading standard. The deviation = where you fell short. The corrective action = what you study harder for next year.' Every student understands controlling instantly when explained through their own exam experience.

2. GPS Navigation analogy

Draw on board: A car with GPS. Planning = the GPS set the route from Mehsana to Ahmedabad. Controlling = the GPS checks every second if the car is on the planned route. When the car takes a wrong turn (deviation), the GPS recalculates (corrective action). Without the plan (route), there is nothing to 'control' against. Without the GPS (control), the plan is useless.

3. Break-even Analysis — live calculation

Give students a real scenario: 'Smit Sir is planning a new commerce batch. Monthly fixed cost = ₹60,000. Fee per student = ₹4,000. Variable cost per student = ₹500.' Ask: How many students must enroll for the batch to break even? Students calculate: BEP = 60,000 / (4,000 - 500) = 60,000 / 3,500 = 17.14 ≈ 18 students. Real, relatable, memorable.

4. Board exam focus areas

- 5 steps of controlling (S-M-C-A-T) — 5/6-mark (appears almost every year)
- Importance of controlling — 5-mark question (common)
- Planning vs Controlling relationship — 4/5-mark (regular)
- Traditional + Modern techniques — 4-mark (explain any 2-3 techniques)
- Break-even analysis with numerical — 3-4-mark numerical/application question
- Case study: identify step/technique — 4-6-mark

5. How to end the chapter + the entire Part A powerfully

"You have now studied all 8 functions of management — Planning, Organising, Staffing, Directing, and Controlling (and within Organising: Delegation and Decentralisation). Together, these 8 chapters describe how EVERY great organisation in the world — from a 5-person startup to TATA Group — actually runs. Every CEO, every manager, every leader uses these exact principles every single day. You didn't just study a textbook — you studied the operating system of human organisations. Use this knowledge well."

End of Chapter 8

"What gets measured gets managed." — Peter Drucker

All 8 chapters done. All of Business Studies — complete.

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